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HUJI Applicative Research Report — Healthcare — September 2026

4 high-potential paper(s) · Healthcare · Monthly · September 2026 · Generated September 01, 2026

Hebrew University researchers are advancing groundbreaking solutions across critical sectors, from human health to global food security. Highlights include a novel oral therapeutic for autoimmune diseases, climate-resilient wheat cultivars, and a universal liquid biopsy platform for complex diagnoses. Additionally, an innovative chemical synthesis method promises to accelerate new drug development. These diverse research opportunities represent significant potential for investment and industry partnerships.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Zvi Granot** — Drug Discovery, Immunology, Clinical
- **Zvi Peleg** — AgriTech, Genomics, Synthetic Biology
- **Nir Friedman** — Diagnostics, Genomics, Clinical
- **Zackaria Nairoukh** — Drug Discovery

HEALTHCARE — RESEARCH HIGHLIGHTS

1. Modulating immune toxicity in ulcerative colitis using a small molecule RAB27A inhibitor. **31/50**

"Oral RAB27A inhibitor shows promise for first-in-class treatment of autoimmune colitis, addressing unmet patient needs."

Main researcher: Zvi Granot

Immunyx Pharma, JBP Minrav Building, Jerusalem, 91120, Israel.; Department of Developmental Biology and Cancer Research, Institute for Medical Research Israel Canada, Hebrew University Medical School, Jerusalem, 91120, Israel.

Published in Inflammatory bowel diseases · 2026 Aug 21

ABOUT THIS RESEARCH

Researchers have developed a small molecule inhibitor targeting the RAB27A protein to reduce harmful immune responses in ulcerative colitis. This approach aims to treat inflammatory bowel disease by controlling the secretion of inflammatory mediators from immune cells.

COMMERCIAL ANGLE

Development of a first-in-class oral therapeutic for autoimmune colitis and other inflammatory disorders driven by RAB27A-mediated exocytosis.

COMMERCIALISATION METRICS

Novelty **6/10**

Targeting RAB27A for immune modulation is a scientifically sound approach, but small molecule inhibition of Rab GTPases is a known pursuit in oncology and immunology, making this an incremental application to UC rather than a groundbreaking discovery.

Commercial Potential **8/10**

Targeting RAB27A with a small molecule offers a highly druggable modality for a significant unmet need in ulcerative colitis, providing a clear path toward a proprietary therapeutic product.

Market Size **8/10**

Ulcerative Colitis is a chronic, high-prevalence condition with a multi-billion dollar global market and significant unmet need for targeted, low-toxicity therapies.

Tech Readiness **3/10**

The title indicates early-stage drug discovery focused on a specific molecular target (RAB27A) for a disease state, which typically represents basic laboratory research (proof-of-concept) rather than a clinical-stage asset.

IP Strength **6/10**

Small molecule inhibitors of specific protein targets like RAB27A are generally highly patentable via composition-of-matter claims. However, without a specific chemical scaffold or novel delivery mechanism described, the defensibility is limited to the target validation.

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2. Antagonistic pleiotropy at the stem solidness 1 locus balances wheat stem architecture and yield components under terminal drought. 31/50

"Breakthrough in wheat genomics enables climate-resilient, high-yield cultivars, vital for global food security."

Main researcher: Zvi Peleg · zvi.peleg@mail.huji.ac.il.

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Published in TAG. Theoretical and applied genetics. Theoretische und angewandte Genetik · 2026 Aug 16

ABOUT THIS RESEARCH

Researchers identified a specific genetic locus (stem solidness 1) in wheat that controls the trade-off between stem structural strength and grain yield. The study explains how adjusting this genetic trait can help wheat plants survive terminal drought without sacrificing overall productivity.

COMMERCIAL ANGLE

Development of climate-resilient, high-yield wheat cultivars through precision breeding or CRISPR gene editing targeting the s1 locus.

COMMERCIALISATION METRICS

Novelty 5/10

The identification of a specific locus controlling stem architecture and its trade-off with yield under drought is a valuable incremental contribution, but follows established paradigms of QTL mapping and antagonistic pleiotropy in crop science.

Commercial Potential 7/10

The identification of a specific locus controlling the trade-off between stem strength and yield under drought provides a direct target for marker-assisted breeding or CRISPR editing in wheat, a global staple crop. While highly relevant for seed commercialization, the score is tempered by the inherent complexity of 'antagonistic pleiotropy' which may require precise tuning rather than a simple binary switch.

Market Size 9/10

Wheat is one of the world's most critical staple crops; a genetic solution to maintain yield under terminal drought addresses a massive global food security market with high industrial demand.

Tech Readiness 3/10

The research focuses on identifying a genetic locus and the biological trade-off (antagonistic pleiotropy) between stem architecture and yield, which is fundamental discovery research (TRL 2-3) rather than a developed seed product or breeding protocol ready for commercial deployment.

IP Strength 7/10

The identification of a specific genetic locus (stem solidness 1) governing a trade-off between architecture and yield provides a clear target for Marker-Assisted Selection (MAS) or CRISPR editing, which is highly patentable in agricultural biotech.

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3. Using Disease-Agnostic Genomic Liquid Biopsy in Complex Diagnoses: Real-Life Example of cfChIP-Seq in a Fever of Unknown Origin. 30/50

"Universal liquid biopsy platform revolutionizes diagnosis of complex, rare diseases, offering rapid identification in critical care."

Main researcher: Nir Friedman

The Lautenberg Center for Immunology and Cancer Research, Faculty of Medicine, and the Rachel and Selim Benin School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel.; The Hebrew University Center for Computational Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published · 2026 Aug 7

ABOUT THIS RESEARCH

Researchers utilized a specialized sequencing technique called cfChIP-Seq to analyze cell-free DNA in the blood to diagnose a complex case of fever of unknown origin. This approach allows for a disease-agnostic search for genomic signatures of inflammation or infection without needing a prior hypothesis.

COMMERCIAL ANGLE

Development of a universal liquid biopsy diagnostic platform for rapid identification of rare or occult diseases in critical care settings.

COMMERCIALISATION METRICS

Novelty 7/10

The application of cfChIP-Seq as a disease-agnostic diagnostic tool for Fever of Unknown Origin represents a novel translation of epigenetic profiling to clinical diagnostics, moving beyond traditional oncology uses. However, the novelty is tempered by the fact that it applies existing ChIP-Seq methodology to a new clinical indication rather than introducing a fundamentally new molecular biology technique.

Commercial Potential 7/10

The use of cfChIP-Seq as a disease-agnostic diagnostic platform for complex cases like FUO suggests high versatility and a clear path toward a high-value clinical service. However, commercial scaling is tempered by the current complexity of ChIP-Seq workflows compared to standard NGS.

Market Size 8/10

The technology is a disease-agnostic platform for complex diagnoses, meaning its addressable market extends beyond a single indication to all undiagnosed systemic illnesses and infectious diseases. This platform-enabling approach significantly maximizes potential market size compared to targeted assays.

Tech Readiness 4/10

The paper presents a 'real-life example' case study, indicating the technology is functional in a clinical setting, but it remains a research-grade application of cfChIP-Seq rather than a standardized, scalable diagnostic product.

IP Strength 4/10

The paper describes the application of cfChIP-Seq, an established genomic method, to a specific clinical use case (FUO) rather than introducing a novel, patentable platform or proprietary molecular tool. Defensibility relies on clinical validation and data rather than a protectable technological breakthrough.

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4. Stereocontrolled entry to borylated piperidines <i>via</i> stepwise dearomative carboboration of pyridines.

29/50

"Novel synthesis method efficiently creates complex drug building blocks, accelerating pharmaceutical research and development."

Main researcher: Zackaria Nairoukh · z.nairoukh@mail.huji.ac.il.

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Published · 2026-08-17

ABOUT THIS RESEARCH

Researchers have developed a new chemical method to transform simple pyridine molecules into complex, functionalized piperidine structures with high precision. This two-step process uses palladium catalysis to ensure the new chemical groups are attached to specific locations and orientations.

COMMERCIAL ANGLE

This method provides a highly efficient way to synthesize complex nitrogen-containing building blocks that are essential for creating new pharmaceutical drugs.

COMMERCIALISATION METRICS

Novelty 7/10

The research presents a highly sophisticated stepwise approach to complex piperidine synthesis and proposes a novel Pd(IV) mechanistic pathway, though dearomative functionalization and carboboration are established themes in organic synthesis.

Commercial Potential 6/10

The methodology offers a highly valuable synthetic tool for creating complex piperidine scaffolds used in drug discovery, but it remains a mechanistic/synthetic improvement rather than a standalone product or platform. Its commercial value is primarily as a licensable process or a contribution to the toolkit of medicinal chemistry outsourcing services.

Market Size 6/10

The method targets substituted piperidines, which are ubiquitous motifs in the pharmaceutical industry, providing access to high-value building blocks. However, as a specific synthetic methodology rather than a platform for a novel therapeutic class, the immediate market is limited to the specialized medicinal chemistry reagent space.

Tech Readiness 3/10

This is fundamental methodology research focused on synthetic organic chemistry and mechanistic validation, representing early-stage laboratory discovery rather than a developed product or platform. While the ability to create substituted piperidines is chemically valuable, the work currently lacks the process optimization or application-specific validation required for higher TRLs.

IP Strength 7/10

The stepwise dearomative carboboration represents a novel synthetic methodology with high potential for patenting diverse chemical space, though its defensibility is tied to the broader scope of palladium-catalyzed dearomatization. The ability to create complex, stereocontrolled piperidines—a privileged scaffold in medicinal chemistry—provides significant industrial value and a strong foundation for protecting specific downstream medicinal applications.

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